A. Very Short

1. What are the advantages of a circular queue over a linear queue?
2. Write any four applications of stack.
3. Find the postfix expression of a + b $ c / d * e.
4. Where can we use the Divide and Conquer algorithm? Write any two examples?
5. What do you mean by Big Oh Notation?
6. What are the main features of queue?
7. What are the main features of stack?
8. Evaluate the postfix expression 2 3 1 * + -
9. What is Rate of Growth?
10. What are recursive algorithm and base case?
11. What are time and space complexity?
12. Define ADT Queue and its operations.
13. . Differentiate between linear and non-linear data structure.
14. What do you mean by Time Space Tradeoff?
15. What is a queue and how does it work?
16. What is ADT?
17. Difference between stack and queue.
18. What is asymptotic notation?
19. What do you mean by run time calculation?
20. What is best/worst case ?

B. Short

1. What is ADT? Explain stack and queue as ADT.
2. How can we analyze an algorithm? Time complexity and space complexity.
3. Why Stack is called LIFO data structure? Write the algorithms and program for push and pop operation on stack using array with full and empty conditions.
4. Why Queue is called FIFO in data structure? Write the algorithms and program for enqueue and dequeue operation on queue using array with full and empty conditions.
5. Explain Recursion and write a C-program to solve TOH for n number of disks.
6. What are the advantages, disadvantages, and applications of linked lists? Explain its types and operations.
7. A) Explain Recursion on brief with suitable example using c.
   B) Write about singular and Double Linked List.
8. Convert the following infix expression to postfix **((A+B)-C*(D/E))+F**
9. Write a recursion algorithm to find Fibonacci Series upto N terms?
10. What is the advantage of postfix notation .Convert the infix expression **a+b*e+(d*e+f)*g** into Prefix expression.
11. What are the advantages and disadvantages of linked lists? Explain any two operations of singly linked list.

C. Case/Long Questions

1. A stack is an Abstract Data Type (ADT), commonly used in most programming languages. It is named stack as it behaves like a real-world stack, for example a deck of cards or a pile of plates, etc. In real-world stack allows operations at one end only. For example, we can place or remove a card or plate from the top of the stack only. Likewise, Stack ADT allows all data operations at one end only. At any given time, we can only access the top element of a stack. This feature makes it LIFO data structure. LIFO stands for Last-in-first-out. Here, the element which is placed (inserted or added) last is accessed first. In stack terminology, insertion operation is called PUSH operation and removal operation is called POP operation.
   i. What is Stack? Explain how the (pop, push, empty, top-of-stack operations are implemented with an example.
   ii. Explain how stack differs with queue and mention some applications of using stack.
2. A doubly linked list is a linear data structure where each node contains a reference to both the previous and next nodes in the list. This allows for efficient traversal in both directions, making it useful in scenarios where forward and backward navigation is required.
   In a doubly linked list, each node holds two pointers: one pointing to the previous node and another pointing to the next node. This bidirectional linkage facilitates operations like insertion and deletion, as it provides easy access to neighboring nodes.
   i How would you insert new node after a specified node?
   ii How would you delete a node after a specified node?
3. A queue is a container of objects (a linear collection) that are inserted and removed according to the first-in first-out (FIFO) principle. An excellent example of a queue is a line of students in the food court of the UC. New additions to a line made to the back of the queue, while removal (or serving) happens in the front. In the queue only two operations are allowed enqueue and dequeue. Enqueue means to insert an item into the back of the queue, dequeue means removing the front item.
   Questions:

a. Explain the operations performed in a queue.

b. Implement enqueue and dequeue operations using the language C for a circular queue.

4. A queue is a container of objects (a linear collection) that are inserted and removed according to the first-in first-out (FIFO) principle. An excellent example of a queue is a line of students in the food court of the UC. New additions to a line made to the back of the queue, while removal (or serving) happens in the front. In the queue only two operations are allowed enqueue and dequeue. Enqueue means to insert an item into the back of the queue, dequeue means removing the front item

Question:

A . Explain the operations performed in a queue.

b. Explain circular queue, double-ended queue and priority queue.

5. Define singly linked list. Write the steps and code used to perform the following operation in a single linked list

a. To insert a new node at the end of the list

b. To remove a first node from the list.

6. Define a singly and double linked list. Write an program for Enqueue and Dequeue operation using linked list of a Queue